

PA2 Report - Requirement 1: User Research and User Analysis

Group: 06

Course: CSC13112 - UI/UX Design

Assignment: Project Assignment 2

Product scope: Freestyle Chess mobile website on smartphone browser

Research methods: Online survey and direct interview

1. Research Overview

1.1 Research Context

In PA1, our group analyzed the Freestyle Chess website and identified several mobile usability issues, including difficult one-handed navigation, long information lists, limited search/filter support, unclear onboarding for first-time users, and unexpected video playback behavior.

In PA2, our group continues with the same product scope and conducts user research to understand how real users interact with the Freestyle Chess mobile website in realistic mobile contexts.

1.2 Research Goals

The user research focuses on achieving the following objectives:

1. **Understand user expectations:** Identify what information mobile chess followers want to access first on a sports/chess event website.
2. **Evaluate homepage clarity:** Assess if first-time mobile users can easily comprehend the core purpose of Freestyle Chess from the homepage.
3. **Assess navigation usability:** Investigate how easily users can navigate between major sections on a smartphone browser.
4. **Test information findability:** Measure the efficiency and discoverability when users look for player ratings, event news, schedules, and video/livestream content.
5. **Identify mobile-specific constraints:** Observe how physical grip (one-handed usage) and short browsing sessions affect the user experience.
6. **Prioritize user pain points:** Gather quantitative and qualitative evidence to prioritize which usability drawbacks from PA1 should be addressed in the redesign proposal.

1.3 Research Questions (Google Form Survey)

The online survey contains 29 questions matching the research goals. Below is the mapping of each question to the related website area and the expected data type:

ID	Research Question (Vietnamese)	Related Area	Data Type / Expected Evidence
Q01	Bạn có thường theo dõi tin tức, giải đấu, kỳ thủ hoặc nội dung liên quan đến cờ vua không?	Screening / Chess Interest	Single choice (Multiple choice)
Q02	Trước khảo sát này, bạn đã từng nghe đến Freestyle Chess chưa?	Screening / Familiarity	Single choice (Multiple choice)
Q03	Bạn thường dùng thiết bị nào để đọc tin tức thể thao hoặc tin tức cờ vua?	Screening / Device Context	Multiple choice (Checkboxes)
Q04	Khi dùng điện thoại, bạn thường cầm máy bằng cách nào?	Screening / Grip Context	Single choice (Multiple choice)
Q05	Bạn thường xem tin tức hoặc thông tin sự kiện trên điện thoại trong tình huống nào?	Screening / Mobile Context	Multiple choice (Checkboxes)
Q06	Khi truy cập một website về cờ vua hoặc giải đấu cờ vua, bạn thường muốn xem thông tin gì trước?	Home page / Expectations	Multiple choice (Checkboxes)
Q07	Nếu chỉ có 2-3 phút để xem nhanh trên điện thoại, bạn muốn website giúp bạn làm gì nhanh nhất?	Home page / Expectations	Short text response
Q08	Theo bạn, điều gì làm một website tin tức/sự kiện trên điện thoại dễ sử dụng?	General Mobile Usability	Short text response
Q09	Khi nhìn trang chủ Freestyle Chess trên điện thoại, bạn có dễ hiểu website này dùng để làm gì không?	Home page / Clarity	Rating scale (1-5)
Q10	Bạn có thấy trang chủ cung cấp đủ hướng dẫn cho người dùng mới không?	Home page / Onboarding	Single choice (Multiple choice)
Q11	Menu điều hướng trên mobile có dễ tìm và dễ bấm không?	Navigation menu	Rating scale (1-5)
Q12	Khi dùng một tay, bạn có thấy các nút điều hướng chính dễ chạm tới không?	Navigation / Reachability	Rating scale (1-5)
Q13	Bạn có từng cảm thấy phải với tay, đổi cách cầm điện thoại hoặc dùng tay còn lại để thao tác không?	Navigation / Reachability	Single choice (Multiple choice)
Q14	Các mục trong menu như News, Schedule, Rating, Videos & Streams có dễ hiểu không?	Navigation / Labels	Rating scale (1-5)
Q15	Bạn có nghĩ việc thêm icon vào các mục menu sẽ giúp nhận biết nhanh hơn không?	Navigation / Labels	Single choice (Multiple choice)
Q16	Việc tìm một kỳ thủ hoặc thông tin trong bảng xếp hạng có dễ không?	Rating page / Info retrieval	Rating scale (1-5)

ID	Research Question (Vietnamese)	Related Area	Data Type / Expected Evidence
Q17	Khi danh sách thông tin quá dài, bạn thích cách hỗ trợ nào nhất?	Rating page / Long lists	Multiple choice (Checkboxes)
Q18	Khi tìm tin tức về một sự kiện cờ vua, bạn muốn tìm theo cách nào?	News page / Info retrieval	Multiple choice (Checkboxes)
Q19	Bạn có từng gặp khó khăn khi không nhớ chính xác tên sự kiện, kỳ thủ hoặc thông tin cần tìm không?	News page / Memory load	Single choice (Multiple choice)
Q20	Khu vực video/livestream có dễ hiểu và dễ sử dụng không?	Videos & Streams page	Rating scale (1-5)
Q21	Khi chọn một video mới, bạn kỳ vọng điều gì xảy ra?	Videos & Streams page	Single choice (Multiple choice)
Q22	Nếu nhiều video phát cùng lúc, trải nghiệm của bạn sẽ như thế nào?	Videos & Streams page	Short text response
Q23	Bạn có từng không chắc một nội dung/nút/thẻ trên website có bấm được hay không?	General Mobile Usability	Single choice (Multiple choice)
Q24	Phần dễ sử dụng nhất của Freestyle Chess mobile web là gì?	Overall Usability	Short text response
Q25	Phần gây khó chịu hoặc khó hiểu nhất là gì?	Overall Usability	Short text response
Q26	Theo bạn, website này phù hợp nhất với nhóm người dùng nào?	Persona / Target Audience	Single choice (Multiple choice)
Q27	Bạn có muốn quay lại website này để theo dõi Freestyle Chess hoặc tin tức cờ vua không?	Return Intention	Single choice (Multiple choice)
Q28	Nếu được cải thiện một điểm trên giao diện mobile, bạn muốn cải thiện điều gì nhất?	General Feedback	Short text response
Q29	Bạn có góp ý nào khác cho nhóm thiết kế không?	General Feedback	Short text response

2. Research Methods

Our group used two user research methods:

1. **Online survey:** Participants opened the Freestyle Chess website on their phone, explored the website for a few minutes, and answered a Google Form based on their experience.
2. **Direct interview:** A smaller number of participants were interviewed directly. During the session, they were asked to perform several tasks on the mobile website while the research team observed their behavior and asked follow-up questions.

This combination helps the group collect both broad and deep data. The online survey provides responses from more users, while direct interviews provide richer qualitative insights, user quotes, and

behavioral evidence.

2.1 Online Survey

Item	Description
Method	Google Form survey
Product tested	Freestyle Chess mobile website
Website URL	https://www.freestyle-chess.com/
Device context	Smartphone browser
Estimated exploration time	3-5 minutes
Estimated form completion time	5-8 minutes
Number of responses	72
Survey period	July 17, 2026 - July 20, 2026
Data collected	Multiple choice answers, checkbox answers, 1-5 rating scales, short text responses
Data source	Form Responses CSV

2.2 Direct Interview

Item	Description
Method	Semi-structured interview with task-based observation
Product tested	Freestyle Chess mobile website
Device context	Participant's smartphone if possible
Number of participants	10 participants (across 13 interview sessions)
Interview duration	10-15 minutes per session
Interview period	July 18, 2026 - July 22, 2026
Data collected	Observation notes, user comments, task difficulties, notable quotes
Current status	Completed

2.3 Why These Methods Were Chosen

The online survey was chosen because it allows the group to gather feedback from many users in a short amount of time. It is useful for identifying common patterns, such as how many users find the navigation difficult or how many users expect search/filter support.

The direct interview was chosen because some usability problems cannot be fully understood through survey answers alone. By watching users interact with the website, the group can observe hesitation,

repeated scrolling, accidental taps, grip changes, and confusion that users may not describe clearly in a form response.

3. Participant Profile

3.1 Participant Summary

Participant Group	Description	Number of Participants
Online survey respondents	Users who tested the website independently and answered the Google Form	72
Direct interview participants	Users who joined task-based interviews with the research team	10 (13 sessions)
Total participants	Combined number of users involved in the research	82

3.2 Chess Interest Level

Chess Interest Level	Number of Users	Percentage	Notes
Very often follow chess news/tournaments	12	16.7%	Core chess followers
Sometimes follow chess content	26	36.1%	Largest group in the sample
Only follow during major events	13	18.1%	Event-driven users
Rarely or never follow chess content	21	29.2%	New or low-interest users

3.3 Familiarity With Freestyle Chess

Familiarity Level	Number of Users	Percentage	Notes
Heard of and visited before	11	15.3%	Existing users
Heard of but never visited	23	31.9%	Aware but inexperienced
Never heard of it before the research	38	52.8%	First-time users

3.4 Mobile Usage Context

Context	Number of Users	Notes
At home	38	Most common mobile browsing context
Before sleeping	26	Supports short, casual browsing sessions

Context	Number of Users	Notes
While commuting, walking, or waiting	25	One-handed use and quick scanning may matter
During a short break	29	Users need fast access to key information
During a live match or livestream	10	Users may need quick supporting information while watching

3.5 Device and Phone Holding Habits

Device Usually Used for Sports/Chess News		Number of Users	Percentage
Smartphone		46	63.9%
Laptop or desktop		41	56.9%
Tablet		12	16.7%
Other		1	1.4%

Phone Holding Habit	Number of Users	Percentage	Interpretation
One hand	32	44.4%	One-handed reachability is a common requirement
Two hands	10	13.9%	A smaller group uses more stable interaction
Depends on situation	30	41.7%	The interface still needs to work across changing mobile contexts

4. Research Procedure

4.1 Online Survey Procedure

Participants were asked to:

1. Open the Freestyle Chess website on a smartphone browser.
2. Explore the website for about 3-5 minutes.
3. Check important sections such as Home, News, Schedule, Rating, and Videos & Streams.
4. Return to the Google Form and answer questions based on their actual experience.

The survey included:

- Screening questions about chess interest and device usage.
- Questions about user expectations when using chess event websites.
- Rating-scale questions about homepage clarity, navigation, information finding, and one-handed use.
- Open-ended questions about frustrations, confusing areas, and suggested improvements.

4.2 Direct Interview Procedure

The direct interviews were conducted using a highly flexible, **semi-structured conversational approach**. Interviewers did not force participants through a rigid, sequence-based checklist of instructions. Instead, they introduced open-ended scenarios and allowed participants to explore the mobile website naturally, adapting specific prompts and tasks in real-time based on each participant's background, hand dominance, and interaction choices.

The team guided the interviews around six **Exploratory Focus Areas** rather than static tasks:

- 1. Homepage Discovery & First Impression:** Participants opened the site and shared their immediate perceptions. Interviewers observed whether the visual hierarchy, text copy, or onboarding cues clearly explained the website's purpose without user confusion.
- 2. Natural Navigation & Menu Access:** Participants were asked to move to main sections like ratings, schedules, or videos. Interviewers observed whether they naturally located the menu button, scanned the text-only labels easily, or faced difficulty.
- 3. Player Information Retrieval:** Participants attempted to find details for a specific player or browse the rating leaderboard. Interviewers observed search behavior, scrolling effort, and how they handled long vertical lists on small screen interfaces.
- 4. Tournament News Filtering:** Participants attempted to find articles about past or current chess events. Interviewers observed how they navigated chronological feeds and dropdown lists without search keywords.
- 5. Media Player Interaction:** Participants selected a video or livestream to watch. Interviewers observed how they expected the player to load, and their reactions to multiple playbacks and controls.
- 6. Physical reachability & Grip Comfort:** Participants were asked to try one-handed scrolling and navigation. Interviewers focused on finger stretching (thumb zone constraints), grip shifts, or accidental taps during single-handed interaction.

Throughout these session activities, the research team documented:

- Critical moments where the participant paused, hesitated, or verbalized confusion.
- Variations in grip styles (one-handed hold vs. two-handed use).
- Spontaneous comments, suggestions, or frustration points.
- Errors or bugs encountered during natural browsing.

5. Online Survey Findings

The online survey collected 72 responses (source data: [Form Responses CSV](#)). The findings below summarize the main quantitative results and selected patterns from short-text responses.

5.1 User Expectations

Expected Information	Number of Users	Percentage	Interpretation
Latest news	38	52.8%	Users expect the website to highlight fresh updates

Expected Information	Number of Users	Percentage	Interpretation
Schedule	22	30.6%	Schedule is useful but not the top priority
Match results	46	63.9%	Most selected information type
Player information	31	43.1%	Users care about player context
Ratings	33	45.8%	Rating information is an important reference need
Videos or livestreams	22	30.6%	Media content matters to a smaller but relevant group
Rules or tournament format	16	22.2%	Newer users may still need explanation

Insight: Users primarily want fast access to match results, latest news, ratings, and player information. In short 2-3 minute mobile sessions, many short-text answers also mentioned "results", "summaries", "time", and "developments", suggesting that users prefer concise, scannable event updates.

5.2 Homepage Clarity

Question	Average Score	Key Interpretation
The website purpose is easy to understand	3.68 / 5	Generally understandable, but not fully clear for all users
The homepage provides enough guidance for new users	26 users said clear; 24 said partly clear; 13 said difficult for new users; 9 were unsure	A large portion still needs more explanation or onboarding

Insight: The homepage is not a complete failure, but it does not fully support first-time users. This matters because 38 out of 72 respondents (52.8%) had never heard of Freestyle Chess before the research.

5.3 Navigation and One-Handed Use

Question	Average Score	Key Interpretation
Mobile navigation menu is easy to find and tap	3.94 / 5	Users can generally find the menu
Main controls are easy to reach with one hand	3.50 / 5	Reachability is weaker than discoverability
Menu labels are easy to understand	3.83 / 5	Labels are mostly understandable, but recognition could improve

One-Handed Usage Issue	Number of Users	Notes
Had to stretch thumb, change phone grip, or use the other hand	31	43.1% reported effort during operation
Did not notice whether this happened	23	31.9% were unsure or did not pay attention
Did not experience this issue	18	25.0% reported no issue

Insight: The menu is visible enough, but physical reachability remains a meaningful mobile issue. Since 32 users usually hold the phone with one hand and 30 users switch depending on context, the interface should support comfortable one-handed operation.

5.4 Finding Information

Question	Average Score	Key Interpretation
Finding a player or rating information is easy	3.60 / 5	Moderately easy, but not highly efficient
Users had difficulty when they could not remember the exact event/player/information name	44 users (61.1%) answered yes	Search and flexible filtering are important
Preferred Support for Long Lists	Number of Users	Interpretation
Sticky search bar	28	Users want search to remain accessible while scrolling
Pagination	28	Users want long lists to be divided into manageable chunks
Filters by event/date/player	31	Most selected support method
Search suggestions	21	Helpful when users do not remember exact names
Only scrolling is enough	13	A minority is comfortable with scrolling only

Insight: Users prefer active support for long information lists. Filters, sticky search, pagination, and search suggestions are all stronger needs than simply relying on scrolling.

5.5 Video and Livestream Experience

Question	Average Score / Count	Interpretation
Video/livestream section is easy to use	3.67 / 5	Usable, but still has expectation and clarity issues

Question	Average Score / Count	Interpretation
Users expect the old video to stop when selecting a new one	38 users (52.8%)	Most users expect familiar single-player behavior
Users were unsure whether some content was clickable	32 users (44.4%)	Clickability needs clearer visual signifiers

Insight: The video section should match familiar media-player mental models. Open-ended responses about multiple videos playing at once often described the experience as "annoying", "distracting", or "not good".

5.6 Return Intention

Answer	Number of Users	Percentage	Interpretation
Yes	17	23.6%	Users are willing to return
Maybe	39	54.2%	Interest exists, but experience must improve
No	4	5.6%	Small rejection group
Not sure	12	16.7%	Users need more value or clarity

Insight: The website has potential retention because 77.8% of users answered "Yes" or "Maybe". However, many are not fully committed, so improving clarity, navigation efficiency, and information finding may increase return intention.

6. Direct Interview Findings

In addition to the survey, our group conducted task-based observation interviews. Each member recorded and reviewed their own participant clips to capture raw user reactions, physical phone handling, and specific usability struggles.

Below are the detailed entries for each participant session, linking directly to the interview recordings and analyzing the key behaviors observed:

6.1 Entry 1: Participant P01 (Reviewed by Le Mai Hoai Bao)

- **Clip Link:** [Google Drive Clip](#)
- **Participant Description:**
 - **Fullname:** M.T.K.D
 - **Gender:** Female
 - **Device used during interview:** Unknown
 - **Description:** New user, does not regularly follow chess news. When using the phone, usually holds it in the left hand and uses the right thumb to scroll and interact.
- **Analysis & Key Observations:**
 - **Interactive Chessboard experience:** When tapping to view a specific event/match, the user saw an interactive chessboard with controls like *Next Step* and *Back Step*. They repeatedly

pressed these buttons to replay match steps.

- *Behavior Analysis & Perception:* This behavior shows that the visual game replay feature on mobile is highly engaging, rather than just reading static text news.
- *Design Implication:* This is a highly valuable feature and should be kept. In mobile redesign, these replay controls must be positioned within the natural thumb zone, with adequate size and spacing to prevent accidental clicks or thumb fatigue.

6.2 Entry 2: Participant P01 (Reviewed by Le Mai Hoai Bao)

- **Clip Link:** [Google Drive Clip](#)
- **Participant Description:**
 - **Fullname:** M.T.K.D
 - **Gender:** Female
 - **Device used during interview:** Unknown
 - **Description:** New user, does not regularly follow chess news. When using the phone, usually holds it in the left hand and uses the right thumb to scroll and interact.
- **Analysis & Key Observations:**
 - **Confirmation Dialog is too small:** When a confirmation dialog appeared (e.g. to confirm actions or choices), the dialog's viewing frame was too small, cutting off the text content and forcing the user to scroll repeatedly to read the whole message before making a decision.
 - *Behavior Analysis & Perception:* Mobile confirmation dialogs need to be short and clear for quick scanning. Forcing the user to scroll inside a tiny pop-up (nested scroll) breaks the flow and causes frustration. The participant suggested expanding the dialog height to display content more naturally.
 - *Design Implication:* When designing mobile confirmation dialogs, keep messages concise and use auto-height dialog containers to avoid scrollbars. Action buttons (Confirm/Cancel) must be prominent and easy to tap for efficient task completion.

6.3 Entry 3: Participant P01 (Reviewed by Le Mai Hoai Bao)

- **Clip Link:** [Google Drive Clip](#)
- **Participant Description:**
 - **Fullname:** M.T.K.D
 - **Gender:** Female
 - **Device used during interview:** Unknown
 - **Description:** New user, does not regularly follow chess news. When using the phone, usually holds it in the left hand and uses the right thumb to scroll and interact.
- **Analysis & Key Observations:**
 - **Navigation Bar is difficult for one-handed use:** The participant feedback indicated that placing the navigation bar in the bottom right corner would make one-handed navigation and operation much easier and more convenient.
 - *Behavior Analysis & Perception:* When holding the phone in one hand and interacting with the other, the natural thumb zone lies in the bottom half of the screen. Placing navigation items in the bottom right corner minimizes finger stretching and grip changes.
 - *Design Implication:* For mobile UI, primary navigation menus or critical action buttons should be placed in the lower half of the screen (e.g. bottom navigation bar or floating

action button at the bottom-right corner) to support one-handed ergonomic reachability.

6.4 Entry 4: Participant P01 (Reviewed by Le Mai Hoai Bao)

- **Clip Link:** [Google Drive Clip](#)
- **Participant Description:**
 - **Fullname:** M.T.K.D
 - **Gender:** Female
 - **Device used during interview:** Unknown
 - **Description:** New user, does not regularly follow chess news. When using the phone, usually holds it in the left hand and uses the right thumb to scroll and interact.
- **Analysis & Key Observations:**
 - **No Loading Indicator during page transitions:** When the user tapped navigation items to switch pages, the system displayed no loading indicator or progress status. This made the user confused, unsure if the website was working or frozen.
 - *Behavior Analysis & Perception:* Without immediate visual feedback from the system (violating System Status Visibility), users assume the website is broken or unresponsive, leading to frustrating repeated taps (rage clicks).
 - *Design Implication:* Add page-load indicators (e.g., loading spinners, progress bars, or smooth transition skeletons) to provide instant visual confirmation that the page is loading when a navigation item is selected.

6.5 Entry 5: Participant P02 (Reviewed by Truong Cong Thien Phu)

- **Clip Link:** [Google Drive Clip](#)
- **Participant Description:**
 - **Fullname:** V.Q.T
 - **Gender:** Male
 - **Device used during interview:** Samsung Galaxy phone
 - **Description:** New user, occasionally watches chess news but has never heard of Freestyle Chess. Frequently uses smartphone to view and read new updates.
- **Analysis & Key Observations:**
 - **Inconvenient when scrolling horizontally (Home Page) (at 00:00):** When V.Q.T scrolled to the "Video & Streams" section, which is a horizontal list, part of the thumbnail was cut off. This visual cue made him expect a swipeable list, but the page does not support horizontal swiping. Instead, it relies on "Prev" and "Next" buttons. This breaks the interface's consistency since other sections support swiping. Using buttons is also unnatural for mobile users, causing them to hesitate and try to figure out why the swipe gesture didn't work.
 - **Menu Button is out of reach:** When starting, V.Q.T held the phone in the right hand while the left hand was idle. The Menu button (three horizontal lines icon) is placed at the top-left, making it impossible to reach with the right thumb in a natural grip. V.Q.T had to shift the fingers and stretch the thumb, barely touching the button. Later, V.Q.T switched the phone to the left hand but still found it hard to reach. Eventually, V.Q.T held the phone with the left hand and tapped with the right index finger, losing one-handed convenience.
 - **Unpredictable Buttons and Icons in Schedule (at 00:28):** When the Menu list opened, V.Q.T selected the "Schedule" tab, which had a down-arrow icon implying content would

expand vertically. However, the information slid in from the left, causing him to pause in confusion for a second. The Schedule view had two close buttons at the top: an "X" and "<- Back to Main Menu". V.Q.T did not understand why there were two ways to exit the view, making it hard to decide. This issue occurs because the Schedule is not a separate page, but a modal/sidebar overlaying the Menu. Tapping "X" closes the entire Menu, while "Back to Main Menu" returns to the navigation list.

- **Search bar (Rating Page) (at 1:51):** When opening the Rating page, V.Q.T scrolled down quickly, then back to the top to use the search bar. He noted that search bars are standard on long listing pages based on experience, and they immediately recognized the magnifying glass icon.
- **Poor Searching Method (News Page) (at 2:11):** The News page has near-infinite content without pagination or progress indicators, making it hard for V.Q.T to find specific items. He used the event dropdown filter, which popped up a long list covering the screen with every past chess event. He had to read through it to select the event, then scroll again to find the article. The lack of a search bar forced the user to perform multiple steps, taking 5-10 seconds per step, which could be much longer for older news.
- **Unable to watch video (Video & Stream page) (at 2:34):** After finding a video, V.Q.T clicked to play it, but there was no visual feedback showing that the video was loading or starting. He tried tapping multiple videos, but none played. He was confused since the homepage video worked. He eventually gave up and left the page, showing that users have no way to recover from this feedback failure.

6.6 Entry 6: Participant P03 (Reviewed by Lam Huu Khanh)

- **Clip Link:** [Google Drive Clip](#)
- **Participant Description:**
 - **Fullname:** L.T.K
 - **Gender:** Male
 - **Device used during interview:** Android phone (Unknown Brand)
 - **Description:** A complete newcomer who has never followed chess or heard of Freestyle Chess. Primarily uses the smartphone to browse social media, with a strong preference for one-handed operation.
- **Analysis & Key Observations:**
 - **Overloaded Hero section lacks onboarding for new users:** Upon accessing the homepage, L.T.K was overwhelmed by the Hero section, which contains too much cluttered information (large banners, tournament images, text, and buttons) without explaining what Freestyle Chess is or what the website is for. A first-time user cannot understand the content or gain a clear first impression, creating a significant entry barrier.
 - *Behavior Analysis & Perception:* The Hero section is critical in orienting and retaining new users. The absence of a concise onboarding message combined with too much event clutter causes cognitive overload and leaves the user disoriented.
 - *Design Implication:* Redesign the Hero section to be minimalist, introducing a brief onboarding message about Freestyle Chess for newcomers, with a prominent call-to-action (CTA) button to guide their next steps.
 - **Difficulty in one-handed use and preference for right-side navigation:** During navigation tasks, L.T.K struggled with one-handed use. When tapping the menu or switching pages, the right thumb could not reach the top-left corner. L.T.K suggested that the navigation menu or

primary controls should be positioned on the right side for easy one-handed reach rather than requiring two hands.

- *Behavior Analysis & Perception:* Due to one-handed social media habits, mobile users naturally place their thumb sweep zone in the lower-right area. Placing navigation controls at the top left forces grip changes or two-handed use, slowing down interaction and increasing drop risk.
- *Design Implication:* Relocate the mobile navigation menu to a bottom bar or a floating navigation drawer on the bottom-right corner to accommodate right-handed users.
- **Video player lacks instant feedback:** When L.T.K tapped to watch a video, the website gave no immediate visual feedback (no loading spinner, no state change on the button) to indicate it was loading. L.T.K had to tap repeatedly and wait a long time before the video played.
 - *Behavior Analysis & Perception:* Lacking system status feedback (System Status Visibility) leads users to think the site is frozen or the button is broken, causing frustrated repeated clicks (rage clicks). This increases loading delay and degrades user trust.
 - *Design Implication:* Integrate loading feedback (such as a skeleton loader or a loading spinner) immediately upon tapping. Optimize API loading speed to resolve video streams in under 1 second.

6.7 Entry 7: Participant P04 (Reviewed by Pham Chi Bao Ninh)

- **Clip Link:** [Youtube video](#)
- **Person's Description:**
 - **Fullname:** Prefer to not disclose
 - **Gender:** Male
 - **Device during interview:** Iphone (Unknown model)
 - **Description:** Right handed, one-handed use. Doesn't regularly watch news of any sports. Doesn't play chess. Usually use phone when free, resting.
- **Analysis & Key Observations:**
 - Main Page (1:10):
 - Prefers to know the leaderboard at first.
 - Being not familiar to chess, most people tend to prefer something that is easier to understand outside of chess, like people. Seeing chess players and their rating is easier to understand that "this person is good" instead of reading some random headline about "Player X played this insane move that checkmated Y".
 - Burger Button (2:20):
 - Finds reaching the button annoying due to one hand - right hand use.
 - He originally used a phone lying on 4 fingers, palm and fingers flat, so the thumb is far from the top left. So their grip is already loose, having to reach up means they have to slide their fingers up and push the phone closer to the thumb, because they aren't gripping the phone, it could slid off the hand. The thumb also has to reach up and looked stiff. Also because they had to push the phone up and to the right for the thumb to reach, that means they are no longer look directly at the screen and just pressing on where the button would probably be in their memory. That means they could miss it, lower the phone to look again, and continue until they actually hit the button.

- Burger Menu tab names (2:54):
 - Easily understand what each tab means quickly.
 - Like most other participants, the words does seem clear and short enough that icons are not needed.
- Rating tab (3:13):
 - Prefers other pages like Schedule or News to also have a search bar
 - Participant was prompted to look for another page with "something to search for" and moved to the Schedule tab, and then the News tab, quickly realizing that only the Rating tab is the only one with a search bar. They found this to be quite annoying because they might still need to reach for something, and they prefer to search it rather than scrolling and finding it.
- Video tab (4:32):
 - Prefer the video to not run simutaniously.
 - Multiple videos running at the same time make it hard to focus on one video. They prefer only the latest one running and the previous one to stop.

6.8 Entry 8: Participant P05 (Reviewed by Pham Chi Bao Ninh)

- **Clip Link:** [Youtube video](#)
- **Person's Description:**
 - **Fullname:** Prefer to not disclose
 - **Gender:** Male
 - **Device during interview:** Iphone (Unknown model)
 - **Description:** Right handed, usually one handed, but context dependent. Watch news of sports they like but not chess related. Doesn't play chess but still know notable things about chess. Does not know about Freestyle Chess. Uses phone when free or don't have anything to do.
- **Analysis & Key Observations:**
 - Main page (1:28):
 - Prefer to have something notable about the chess that the page is showing or something that can describe the chess form. Said that if someone knew Freestyle chess already would understand, but the opposite person might not understand anything.
 - This is quite common for most participants in my group. They all would like the home page to show more than just the news. If they already know chess, they would like to know what this chess is about, or that it's just the name of the website and therefore they don't need to worry about. And this page is very much designed for someone who already know what they are getting themselves into, so that is a reasonable observation.
 - Burger Menu (3:23):
 - Feels natural when pressing because the person was using both hands. But when prompted to use one hand, it very much felt uncomfortable. Would like to have it be a bar to drag instead of a button.
 - This is quite normal, if the phone is small enough and the user is either left handed or just happen to have the left hand there, they can easily reach the button as it is on the left corner. The phone being quite small also helped with no needing to reach up with

the left hand. But the right hand still needed that finger pushing + thumb reach to touch.

- Burger menu tab names (4:12):
 - Can easily understand, not much worth talking about.
- Rating tab (4:30):
 - Noted that the word font style look too similar on each row so they might accidentally miss something they were looking for. On the infinite page scroll, they have never seen much of the like, but they prefer it to pagination because it reduce load time and having to press buttons to change pages, but doesn't have any strong opinion due to seeing it so infrequently.
 - This is the only person to have commented on the design of each row looking too monotonous, which could hinder search speed because it forces you to slow down while searching to look. And combined with the phone having a smaller font size and the user is often quicker to do interactions on a phone, it is easy and understandable that they might scroll past something they need completely unintentionally. As for the page scroll list, they actually prefer it, phone interaction usually need to feel quick, users don't want to wait, and being able to school quite a time before need to press a button to scroll more reduces loading time and let users do more things, even if those things could contain problems, like scrolling cross something important.
- Video tab (6:06):
 - Does not like multiple videos running at the same time. Brought up that they might be using too much data if they are on 5G with multiple videos running, implying they do not think they could stop a video once it has started.
 - The fact that they don't like multiple video running is nothing new, but they implied that they couldn't stop a video once it has started running. This feeds into usability and showing the users what they could or could not do. With each video screen being so small, they think it is a stabic embed with no controls, so they think once a video starts, they can't stop it and therefore brought up that they might lose data losing them all.

6.9 Entry 9: Participant P06 (Reviewed by Pham Chi Bao Ninh)

- **Clip Link:** [Youtube video](#)
- **Person's Description:**
 - **Fullname:** Prefer to not disclose
 - **Gender:** Male
 - **Device during interview:** Oppo A9 2020 , Android
 - **Description:** Right handed, 2 hand use most of the time. Read a lot of news, like football. Sometimes read about chess. Plays chess, also knows about Freestyle Chess, but just heard of it, nothing specific.
- **Analysis & Key Observations:**
 - Main page (1:11):
 - Thinks the "Freestyle" is just a name, otherwise feels like it's just a normal chess website that serve chess news.
 - This is the only person in the group that plays chess actively, so they have heard of it before, but they only focus on the main stream chess stance, rather than any subcategories. This is normal, even so the name isn't one, Freestyle Chess is also

known as Chess960. Some people might know the format but not know this site is referring to that.

- Burger menu (2:06):
 - Does not feel any discomfort when needing to press that button. Claims they are used to it due to using many sites with similar button placements, as well as being comfortable with their own phone.
 - This is the only user that reported absolutely no discomfort to the button placement, but not because the button was placed well, but rather a combination of their habits and phone size that they managed to reach it without issues, for someone with a bigger phone they might have problems with it.
- Burger menu tab names (2:28):
 - Also doesn't think that the names are difficult to understand, but did suggest combining some tabs to make it shorter.
 - Again nothing of note here for easily understanding the tab names, the suggestion also seem to be a commonly reached one because some tabs to give names that naturally push people to group them.
- Rating tab search (2:57):
 - User tried to search for someone directly and could not find them, and then later when scrolling down do find their name. This signal that the search bar demand accurate inputs to work. Which they did not like and say that they usually search people by first name last name, instead of last name first name like the website.
 - This is a big issue, the function is clearly hiding information it clear has, and while not its fault that the function didn't return the information since the search term was incorrect, users prefer it when the mistakes is clearer, typo is a very human mistake and easy to overlook, as well as when the website isn't doing its job, people tend to blame the website, rather than thinking they did something wrong
- Rating tab list (3:58):
 - Prefer pagination, does not like to scroll down repeatedly, think that its quicker to jump between pages and see topX people than to scroll.
 - This is among the few people that prefer pagination, because on a phone, pagination is often annoying because the contents have to be small and therefore many pages are needed to display the same amount of information. Their reasoning is that since if a page size is fixed, the page change button is always at the bottom, and then only need to press the same spot repeated to hop pages, which to them is easier.
- News tab (4:33):
 - Wants a search bar, mainly for lossy search, because they might not remember the full name of a competition or a news segment, think that a search bar is both quicker and easier to use than a filter for past events.
 - They prefer a search bar for speed rather than looking from what they are trying to say, since the events are clearly listed in a filter list they could look through, so they don't need to remember as much already if there wasnt anything at all. This is an interesting comment by the user, because usually most people prefer a search bar to search, but they prefer it for the speed, which is a factor too, but usually people think of searching when a search bar is mentioned.
- Video tab (5:25):

- Does not mind multiple video running, this is the only person in my group to actually prefer multiple video to run simultaneously.
- Reasoning given is that they could scroll down and watch a few seconds of the new video to see if it is interesting enough to switch, and if they do decide to, they could go back and switch it themselves. This is rather a special case from many, since most would prefer only one video to run at a time, so they aren't overwhelmed and confused on what they are looking or hearing. This could only be explained by habit, if someone regularly watches multiple things together, they might even prefer the page have this feature, otherwise this is an edge case not worth discussing until more data is gathered.

6.10 Entry 10: Participant P07 (Reviewed by Pham Chi Bao Ninh)

Clip Link: [Youtube video](#)

- **Person's Description:**

- **Fullname:** Prefer to not disclose
- **Gender:** Male
- **Device during interview:** Iphone 12 Pro Max
- **Description:** Right Handed, One handed. Doesn't find news directly, read them accidentally as they look online. Aren't interested in Chess.

- **Analysis & Key Observations:**

- Main page (1:10):
 - Wants to see News segment that quite major, like new records or upcoming tournaments. Thinks that the website is hard to tell what it is trying to show other than chess in general, due to the logo and naming.
 - Nothing of note from the others.
- Burger menu (2:31):
 - Fine with reaching up to the button
 - Again because they are used to it, not because it is actually easy. In the video they are still shown to perform the motion of pushing the phone up to the right then reach with their thumb, despite having a high posture of the hand, because the phone's bottom was resting on the table, the hand was naturally high. Despite that they still had to perform that motion.
- Rating tab (3:10):
 - Prefer scrolling infinitely rather than pagination.
 - Again, like the majority of my group, they all seem to prefer scrolling on the phone. Again this may simply just because the phone naturally encourage such motion, scrolling is quite a standard thing to do on a phone, and pagination might just slow thing down because it needs to load more per segment of time spent searching, but again, depending on the user, that searching might not be fully efficient.
- News tab (4:17):
 - Prefers a search bar to exist, but says nothing beyond "it would be easier"
 - This is rather vague, it could just be that most people prefer to have a search bar just be there and not need it than to need it and not have one. The filter alone is quite good already as it groups by events, and news segment usually concern a short period of time where an event is happening, but a search bar is a common want.

- Video tab (4:45):
 - Want the previous the video to stop itself.
 - Nothing of note

6.11 Entry 11: Participant P08 (Reviewed by Pham Chi Bao Ninh)

Clip Link: [Youtube video](#)

- **Person's Description:**

- **Fullname:** Prefer to not disclose
- **Gender:** Male
- **Device during interview:** Iphone 11 (Unknown Specific Model)
- **Description:** Right Handed, One handed. Doesn't find news directly. Aren't interested in Chess.

- **Analysis & Key Observations::**

- Main page (0:51):
 - Want the to see the people in the chess world first, like who recently won. Thinks the page is not quite enough at explaining itself, like tutorial how to play Freestyle Chess.
 - Again a common opinion, but they don't know that by looking at the front page, they don't see the explanation for the website. They are inside the appropriately named "Rules and FC Players Club" tab, in there the specific rules are inside PDF's and a visual is explain what the game mode is. This is fine but for most users, they want to know immediately what they are doing, so they don't get loss, as they are new and can easily quit if they don't know what they are supposed to be doing and there is no one there to guide them.
- Burger menu (1:55):
 - User felt like they had to reach a little when pressing it.
 - The user says they needed to reach, but visually, their hands are quite big compared to their phone. They are the only person to be able to reach the button without needing to push the phone up at all. Now this could either be habit to keep the screen in view, or because their hands are big, or because this motion is learnt well, all of that I do not know. At first they used 2 hands, the left to hold and the right to press, which is common for right handed people, some though might reach up with their left thumb because then they don't need to move their entire hand. Which is actually seen afterwards when they are prompted to move to other tabs, they sometimes use their left hand, sometimes their right, whichever is in position to be easier to use.
- Rating tab (2:38):
 - Prefers pagination
 - Reasoning given is that the further down they go in the list, it is easier to keep track where they are with pagination than with scrolling infinitely. This often isn't an issue for most searching functions, like google. You rarely cross over the 2nd page because the algorithm is so efficient. But here you are searching for chess players sorted by their rating, which means you might follow someone that isn't highly rated which could take a while. This search bar helps with this of course, but as shown before, if you don't accurately remember their name, the record might not show up.
- News tab (4:00):
 - Doesn't mind the lack of a search bar.

- Among the few that don't really need a search bar and is content with the filter. This segment is short though, they didn't have much to say, it could just be that they don't read news by looking for it that much, but rather just read whatever comes up first, which is how people usually read news, they find a source supplier they like and read the top post of the day, so in that sense, a search function is not really needed for these type of people.
- Video tab (4:31):
 - Prefers the previous video to stop itself.
 - Nothing of note

6.12 Entry 12: Participant P09 (Reviewed by Phung Ngoc Tuan)

- **Person's Description:**

- **Gender:** Male
- **Device used during interview:** Xiaomi POCO F4 phone, Chrome browser
- **Description:** A chess-following user who is familiar with chess content and browses the site as someone who already has an interest in tournaments, players, and match updates. Using the phone with one hand holding the device and the other hand for interaction.

- **Analysis & Key Observations:**

1. **Ranking search requires exact text matching:** [Video clip](#)

The participant tried to find a player by typing "Le Quang", then "Liem Le", and later "Liem". The name only appeared when he entered "Liem", suggesting that the search function depends heavily on the original stored text format (the player's full name is stored with a comma, e.g. "Le, Quang Liem") rather than matching flexibly on partial or reordered name fragments. This made it difficult for the user to search naturally by the player name he already knew, and he did not appear to notice why his first two attempts failed — he simply kept trying different word orders until one worked.

2. **Schedule does not provide enough structure:** [Video clip](#)

In the Schedule section, the participant was specifically looking for a detailed match schedule and a visual tournament bracket, so he could see which matches were happening and who was playing whom in the next round. When the Schedule page did not offer this, he went to the tournament's own information page and even tapped a link leading to an external site, but still could not find the schedule information he wanted. This shows the gap was not a one-off misunderstanding — the user actively searched across multiple parts of the site before giving up, indicating a real, unmet information need rather than a minor inconvenience.

3. **Videos look interactive but do not respond:** [Video clip](#)

The participant tapped the same video thumbnail multiple times in a row expecting it to play, but nothing happened. The first video on the page did autoplay its visuals, giving the impression that the video section was functional, which made the failure of the other videos more confusing rather than less. Because there was no visible feedback or error message, the participant verbally attributed the problem to a weak network connection, when in fact the issue was with the interactivity of the video elements themselves. This is a meaningful

finding: the absence of feedback led the user to form an incorrect mental model of the cause, which could discourage him from retrying later even if the network were fine.

4. Carousel playback causes confusion and audio overlap: [Video clip](#)

On the homepage carousel, the participant could swipe left/right on the *upcoming* video's thumbnail to move the carousel, but once a video started playing, swiping directly on the *currently playing* video no longer worked, since the swipe gesture was now being captured by the video player itself instead of the carousel. To work around this, the participant had to rely on the "Prev"/"Next" buttons instead of the natural swipe gesture he had used everywhere else on the page. More critically, when he moved to a new video and pressed play, the previous video kept playing in the background, causing **two audio tracks to overlap**. The participant noticed this immediately, since two voices were playing at once. This indicates the carousel does not properly stop or pause previous media when the user moves on, which directly breaks the listening experience.

- **Summary**

Overall, the participant had trouble with three main areas: search precision in Ranking, unclear/missing structure in Schedule, and weak interaction behavior in the video sections. The most serious issue was the video experience: not only did tapping fail to produce any visible response — leading the user to wrongly blame the network — but switching videos in the carousel also caused overlapping audio, which was noticed immediately and made the experience feel broken and poorly controlled.

6.13 Entry 13: Participant P10 (Reviewed by Phung Ngoc Tuan)

- **Person's Description:**

- **Gender:** Female
- **Device used during interview:** iPad in landscape mode, placed on a stand; used Apple Pencil for interaction, and sometimes rotated the pencil to tap directly with the same hand. Safari browser.
- **Description:** A newcomer to chess who does not follow the chess scene and has little familiarity with chess-related concepts. She was exploring the site out of general curiosity, trying to understand what content and features it offers.

- **Analysis & Key Observations:**

1. **Hero section video "Stop" button does not actually stop playback: [Video clip](#)**

In the hero section at the top of the homepage, which lists featured news and videos, the participant tapped the "Play video" button under one of the videos. The video started playing and the button correctly changed to "Stop video." However, when she tapped "Stop," the video kept playing while the button still switched back to "Play." She tapped it once more, but the result was the same — the button toggled its label between "Play" and "Stop" while the video itself never actually paused through that control. After this single retry, she reacted with a puzzled "**huh**" and gave up on using the button, simply moving on to explore other parts of the page while the video continued playing in the background. Notably, the video could still be paused by tapping directly on the underlying YouTube player itself, meaning the playback control was not fundamentally broken — only the custom "Play/Stop" button was

disconnected from the actual player state, which is a misleading and confusing interaction for a user who has no reason to expect a separate, "hidden" way to control the video.

2. **Returning to home reveals multiple overlapping, silently-playing videos and a broken "Stop" that resets rather than pauses: [Video clip](#)**

After browsing other sections of the site, the participant came back to the homepage and tapped "Play" on additional videos in the hero section. By this point, three videos ended up playing at the same time. The audio was muted this time (the embedded video players appeared to autoplay in a muted state), so the overlapping playback was silent rather than noisy — but the underlying interaction problem was the same. She tried pressing the "Stop" button on the first video again, and once more it failed to pause the video through that control; the button flipped to "Play," and when she pressed it, the video restarted from the beginning instead of resuming. She repeated this a few times and got the same broken result each time before giving up on the button and moving on. As with the first attempt, the video could still be controlled by interacting with the player itself directly rather than through the custom button. This confirms the issue is a genuine, repeatable bug in the custom control specifically — not in video playback overall — and shows it persists across multiple sessions and even after navigating away and back.

3. **Ranking search does not support searching by title or rating, despite being the information the user actually wanted: [Video clip](#)**

In the Ranking section, the participant tried searching using a chess title abbreviation, typing "FM", which returned only players whose *names* happened to contain the letters "fm," rather than players holding the FM (FIDE Master) title. She also tried entering a numeric rating value, which returned no results at all, since no player's name contains that number. Notably, the search bar was labeled "Search player," but she attempted these searches anyway because there was no other visible way to filter by title or rating. After the searches failed, she did not try to manually scroll through the list to find players by rating; she simply moved on to another part of the site. This indicates a real, unmet expectation — as a newcomer unfamiliar with player names, filtering by title or rating would have been a more natural way for her to explore the rankings than searching by a name she didn't know.

4. **Schedule menu banners appear tappable but aren't, and the menu overlay itself has a broken scroll boundary: [Video clip](#)**

In the Schedule section's menu, two large banners displaying information about two tournaments appeared visually prominent enough that the participant expected them to be tappable links; she tapped on them four times across the two banners, but nothing happened. Separately, she also encountered a scrolling issue with the menu itself: when she scrolled down as far as possible, part of the menu's text remained cut off at the bottom of the screen. She could briefly reveal the hidden text by dragging further and holding, but as soon as she released, the view snapped back and the text disappeared off-screen again. She commented "**this can't be scrolled all the way**" and gave up trying to read the remaining content. This behavior resembles the native browser "overscroll" bounce-back effect, suggesting the overlay's actual content is taller than its visible container, causing part of it to always fall outside the viewable area regardless of how the user scrolls.

- **Summary**

Overall, the participant — a complete newcomer to chess — ran into problems in three main areas:

a misleading custom video control in the homepage hero section, a Ranking search that only supports name matching despite users naturally wanting to filter by title or rating, and a Schedule menu with both non-functional banners and a scroll boundary bug that permanently hides part of its content. The most persistent issue was the hero video's "Play/Stop" button: its visual state never matched the actual state of the video, and this happened consistently across multiple attempts and even after navigating away and back. Importantly, the video itself was not unresponsive — it could still be paused by interacting with the player directly — which means the real problem is a misleading custom control rather than a fully broken feature, but this distinction was invisible to the user, who had no way of knowing there was an alternative way to pause it.

7. Key User Insights

Based on the online survey and direct interview results, our group identified the following key insights. These insights have been refined and validated across both research methods.

Insight 1: Users Prefer Fast Access to Results and Key Event Updates

Finding: Users usually visit a chess event website to quickly check match results, latest news, ratings, and player information.

Evidence: 46 users (63.9%) selected match results, 38 users (52.8%) selected latest news, 33 users (45.8%) selected ratings, and 31 users (43.1%) selected player information. Many short-text answers for 2-3 minute browsing sessions also mentioned results, summaries, time, and important developments.

Meaning: The mobile homepage should prioritize quick, scannable information rather than forcing users to browse deeply before finding useful updates.

Insight 2: First-Time Users Need Better Homepage Explanation

Finding: The homepage is generally understandable, but many users still need clearer guidance about what Freestyle Chess is and what the website helps them do.

Evidence: The homepage clarity score was 3.68/5. Only 26 users (36.1%) said the homepage guidance was clear, while 24 users (33.3%) said it was only partly clear and 13 users (18.1%) said it was difficult for new users. In addition, 38 users (52.8%) had never heard of Freestyle Chess before this research.

Meaning: A stronger first-screen explanation or onboarding cue would reduce confusion for users who arrive without prior knowledge.

Insight 3: Mobile Navigation Is Discoverable but Not Fully Comfortable

Finding: Users can generally find and understand the mobile menu, but one-handed reachability still creates effort.

Evidence: The mobile menu received 3.94/5 for being easy to find and tap, while one-handed reachability received a lower score of 3.50/5. 31 users (43.1%) reported having to stretch, change grip, or use the other hand.

Meaning: Navigation improvements should focus not only on visibility but also on thumb reach, physical comfort, and reducing grip changes.

Insight 4: Users Need Search and Filtering Support for Long or Uncertain Information Tasks

Finding: Users often need help when browsing long lists or when they do not remember exact event/player names.

Evidence: 44 users (61.1%) reported difficulty when they could not remember exact names or information. For long lists, 31 users selected filters, 28 selected pagination, 28 selected sticky search, and 21 selected search suggestions. Only 13 users selected scrolling as enough.

Meaning: The website should support flexible search, event/date/player filters, and list organization to reduce memory load and scrolling effort.

Insight 5: Video Interaction Should Follow Familiar Single-Player Behavior

Finding: Users expect video selection to behave like common media platforms, where selecting a new video stops or replaces the old one.

Evidence: 38 users (52.8%) expected the old video to stop and the new video to play. Open-ended responses about multiple videos playing at once included negative reactions such as "annoying", "distracting", and "not good".

Meaning: The video/livestream section should clearly indicate the active video and prevent multiple videos from competing for the user's attention.

8. Pain Points

Pain Point	User Group Affected	Severity	Evidence	Related PA1 Drawback
Mobile menu is not fully comfortable for one-handed use.	Mobile users, especially users browsing while waiting, walking, or lying down	High	One-handed reachability scored 3.50/5; 31 users (43.1%) reported stretching, changing grip, or using another hand	Low reachability to touch collapsed navigation button
Long lists require better browsing support.	Users finding player/rating information	High	31 users wanted filters; 28 wanted pagination; 28 wanted sticky search; only 13 said scrolling alone was enough	Infinite Rating List
Navigation labels could be recognized faster with stronger visual cues.	New users and casual users	Medium	35 users (48.6%) said icons would help; 24 users (33.3%) were unsure	No Icons in the Navigation Bar
Homepage does not explain the website purpose clearly	First-time users	High	38 users (52.8%) had never heard of Freestyle Chess; homepage clarity was 3.68/5; 37 users wanted more guidance, were unsure, or found it difficult	Home Tab Provides No Onboarding

Pain Point	User Group Affected	Severity	Evidence	Related PA1 Drawback
enough for new users.				
Video selection behavior may not match user expectations.	Video/livestream viewers	Medium	38 users (52.8%) expected the old video to stop when selecting a new one; open responses described multiple videos as uncomfortable or distracting	Playing Multiple Videos Simultaneously
Users need better search/filter support when they cannot remember exact names.	Users finding news, events, ratings, or videos	High	44 users (61.1%) had difficulty when they could not remember exact event/player/information names	No Search Function Across Information Lists

9. Personas

The following personas are fictional user profiles created based on the research findings. They represent major user types observed in the survey and interviews.

Persona 1: Frequent Chess Follower

Attribute	Description
Name	N.H.A
Age	24
Background	Regularly follows chess tournaments, players, and FIDE ratings. Updates daily.
Goals	Quickly check latest match results, player leaderboards, next round schedules, and event news.
Mobile context	Uses the phone during short breaks, while commuting, or during live streams of games.
Needs	Fast navigation, efficient search function, advanced filters (by rating, title), and constantly updated info.
Pain points	Too much scrolling on long lists, sub-optimal search (requiring exact spelling), and unclear schedule layouts.
Quote	"I want to open the site and see the latest results or ratings immediately, not scroll endlessly or type name characters perfectly."

Persona 2: Event-Driven Viewer

Attribute	Description
Name	T.M.P
Age	29
Background	Only follows chess during major events or when friends share match links.
Goals	Find event highlights, general schedule overview, and key player updates.
Mobile context	Uses phone for short browsing sessions (2-3 mins) during free time or before sleeping.
Needs	Simple and intuitive homepage, event-based categories, easy explanations, and a stable video player.
Pain points	Disoriented on where to start, hard to search related news due to lack of keyword search, and annoyed by overlapping video audios.
Quote	"I just want to browse for 2-3 minutes to check the schedules and highlights; the interface should be clear for casual users."

Persona 3: First-Time Mobile User

Attribute	Description
Name	L.T.K
Age	21
Background	A complete newcomer whose mobile habit is mostly social media browsing. Has never followed chess or heard of the website.
Goals	Quickly grasp the website's purpose, play livestream videos without feedback delay, and navigate easily with one hand.
Mobile context	Uses an Android phone with one hand while on the move or relaxing.
Needs	A simple home screen with onboarding guidance, navigation buttons placed in a convenient bottom/right position, and instant video player feedback.
Pain points	An overloaded Hero section causing cognitive fatigue; a hamburger menu placed too high to reach with one thumb; unresponsive video player when played.
Quote	"The homepage has too much text and is confusing; the menu is too far to tap with my thumb, and the video stays frozen for a long time before playing."

10. Work Models

10.1 Flow Model: Finding Player Rating

Step	User Action	System/Interface Response	Problem Observed	Opportunity
1	User opens Freestyle Chess website on phone	Home page appears	Hero section is cluttered with too many banners/news and lacks onboarding information explaining the format to newcomers (P03)	Redesign a minimalist Hero section, adding a brief onboarding/intro banner with a clear CTA
2	User looks for navigation menu	Menu icon is placed on mobile layout	Hamburger button is placed at the top-left, out of reach for one-handed thumb interaction (P01, P02, P03)	Move the menu button to the bottom (Bottom Navigation) or shift it towards the bottom-right corner
3	User opens menu and selects Rating	Rating page appears	Menu slides in as a side panel overlaying the screen, showing sluggish response and long category labels	Design a lightweight menu using tabs or a smooth drawer, responding instantly in under 0.3s
4	User scans or searches for a player	Rating list/search field appears	Leaderboard list is too long (Infinite Scroll) with no sticky search bar, causing repetitive scrolling and losing position context (P01, P02)	Integrate a sticky search bar at the top of the page and use pagination to reduce vertical length
5	User reads player rating information	Player name, title, rating are shown	Shows only plain monotone text, lacking visual elements such as nationality flags or player photos	Add country flags as visual icons and design player cards to improve scanability

10.2 Sequence Model: Finding Event News

Trigger	User Intention	Action	Breakdown / Difficulty	Design Implication
User wants to follow a chess event	Find relevant news quickly	Opens News page	Must open the hamburger menu hidden at the top left, making one-handed access very difficult (P01, P03, P05)	Add the 'News' link to a persistent Bottom Navigation bar for direct thumb access

Trigger	User Intention	Action	Breakdown / Difficulty	Design Implication
News list appears	Identify relevant event	Scrolls or looks for filter	News list displays chronologically in an infinite scroll, lacking clear event filters, forcing users to browse aimlessly (P04)	Add quick filter chips for major events (e.g., 'Geller Cup', 'Freestyle Chess 2026') at the top of the News page
User does not remember exact event name	Search by approximate information	Uses keyword/date/player if available	No search bar to search news by keywords, making it difficult to find older news articles (P04)	Provide a keyword search bar with auto-suggestions for player or event names
User finds article	Read summary or full content	Taps article card	News cards lack interaction feedback (no color change on hover/tap), and click boundaries are ambiguous	Design news cards to provide subtle visual feedback (e.g. background color change on tap) to confirm click state

10.3 Artifact Model: Current Mobile Website Elements

Artifact	Purpose	Current Issue	Evidence	Possible Improvement
Mobile navigation menu	Move between main sections	Hamburger menu is placed too high on the left, hard to reach with one thumb during one-handed use	P01, P03 had to stretch their thumbs completely and almost dropped the phone; P02 changed grips repeatedly	Change to a Bottom Navigation Bar or shift the navigation button to the bottom-right corner
Homepage content	Introduce website and key content	Hero section is overloaded with too much info, lacking onboarding to explain Freestyle Chess to newcomers	P03 (L.T.K) felt overwhelmed, unimpressed, and did not understand the website's purpose as a newcomer	Simplify the Hero section, adding a brief introductory banner with a CTA (e.g., 'Explore Freestyle Chess')
Rating list	Show player rankings	Rating list is too long (Infinite Scroll) with no	P01 scrolled continuously for over 20s; P02	Integrate a sticky search bar at the top of the rating list

Artifact	Purpose	Current Issue	Evidence	Possible Improvement
		sticky search bar and lacking pagination	scrolled up and down to find a player's position	and apply pagination
Search/filter controls	Help users find information	No search bar for news, and the event filter is long and hidden	P04 had to use browser's Ctrl+F to find articles; P02 was confused by the long event list overlaying the screen	Provide a keyword search bar on News/Videos pages and display event filters as selectable tabs
Video/livestream cards	Let users select media content	Tapping play lacks immediate visual feedback, feeling frozen; allowing simultaneous playback causes overlapping audio	P03 clicked repeatedly due to lack of immediate feedback; P02 experienced overlapping audio from two videos, causing frustration	Add a skeleton loader/spinner when playing, and enforce a single-player media model (new play stops previous video)

11. Summary of Findings

Based on the online survey, users generally need the Freestyle Chess mobile website to support short, goal-driven browsing sessions. The most important information includes match results, latest news, player ratings, player information, schedules, and videos/livestreams.

The research suggests that the main mobile UX issues are:

1. Users need faster access to match results and important event updates.
2. First-time users need clearer explanation of what Freestyle Chess is and what the website provides.
3. The navigation menu is discoverable, but one-handed reachability still creates physical effort.
4. Long lists and uncertain search tasks require filters, sticky search, pagination, and search suggestions.
5. Video/livestream interactions should follow familiar single-player media behavior.

These findings have been consolidated and validated based on both the online survey and direct interview sessions. They will be used in Requirement 2 to brainstorm user problems, organize them into an affinity diagram, and generate potential design ideas.

12. Research Limitations

This research has the following limitations:

1. The online survey depends on participants' self-reported answers, so it may not capture every real interaction difficulty.

2. The direct interview sample size is limited to 10 participants (13 sessions) due to time constraints.
3. Some participants may not be familiar with chess websites, which can affect how they interpret the website content.
4. Participants used different phone models and screen sizes, which may affect navigation and reachability feedback.

Despite these limitations, combining survey data with direct interview observations helps the group identify common user needs and mobile usability problems with enough evidence for the next stages of PA2.